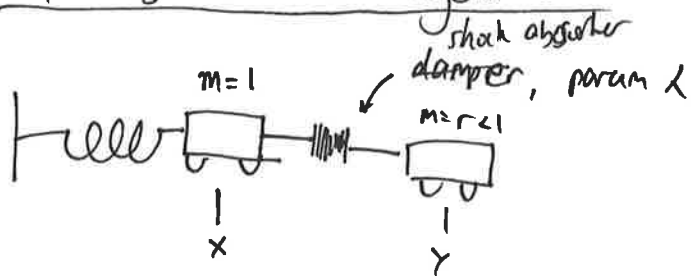


What is a stiff system?



photon pressure $\rightarrow d\gamma$ $\xleftrightarrow{\sim \sigma_T x_e} d\delta$

$$\dot{x} = v$$

$$\dot{y} = w$$

$$v' = -\omega^2 x - \lambda(v - w)$$

$$w' = \frac{1}{r} \lambda(v - w) \leftarrow \text{momentum conservation}$$

Two timescales

T_{slow} : when ports settled into joint motion

$$\sim \frac{1}{\omega} \sqrt{1+r}$$

T_{fast} : how ~~fast~~ quickly ~~damper~~ rel vel damps

$$s = v - w$$

$$s' = -\omega^2 x - \lambda \left(1 + \frac{1}{r}\right) s$$

$$T_{\text{fast}} \sim \frac{1}{\lambda(1 + \frac{1}{r})}$$

λ huge (photon-e scatt happens very often)

MM

When solving, want to ~~resolve both~~ damping and see slow

$$\Delta t < T_{\text{fast}}$$

$$N_{\text{step}} * \Delta t > T_{\text{slow}}$$

$$\Rightarrow N_{\text{step}} \gtrsim \frac{T_{\text{slow}}}{T_{\text{fast}}} \sim \lambda \leftarrow \text{more often } \sigma\text{-e scatter need more time steps.}$$

Insted, TCA: $\omega \rightarrow \nu$ intensely (baryon pinned to σ immediately)
 $S \rightarrow 0$

$$V' + r\omega' = -\omega^2 X \Rightarrow (1+r)V' = -\omega^2 X$$

$$\Rightarrow \text{reduced } \omega_{\text{eff}} \sim \frac{\omega}{\sqrt{1+r}} \quad \text{TCA}$$

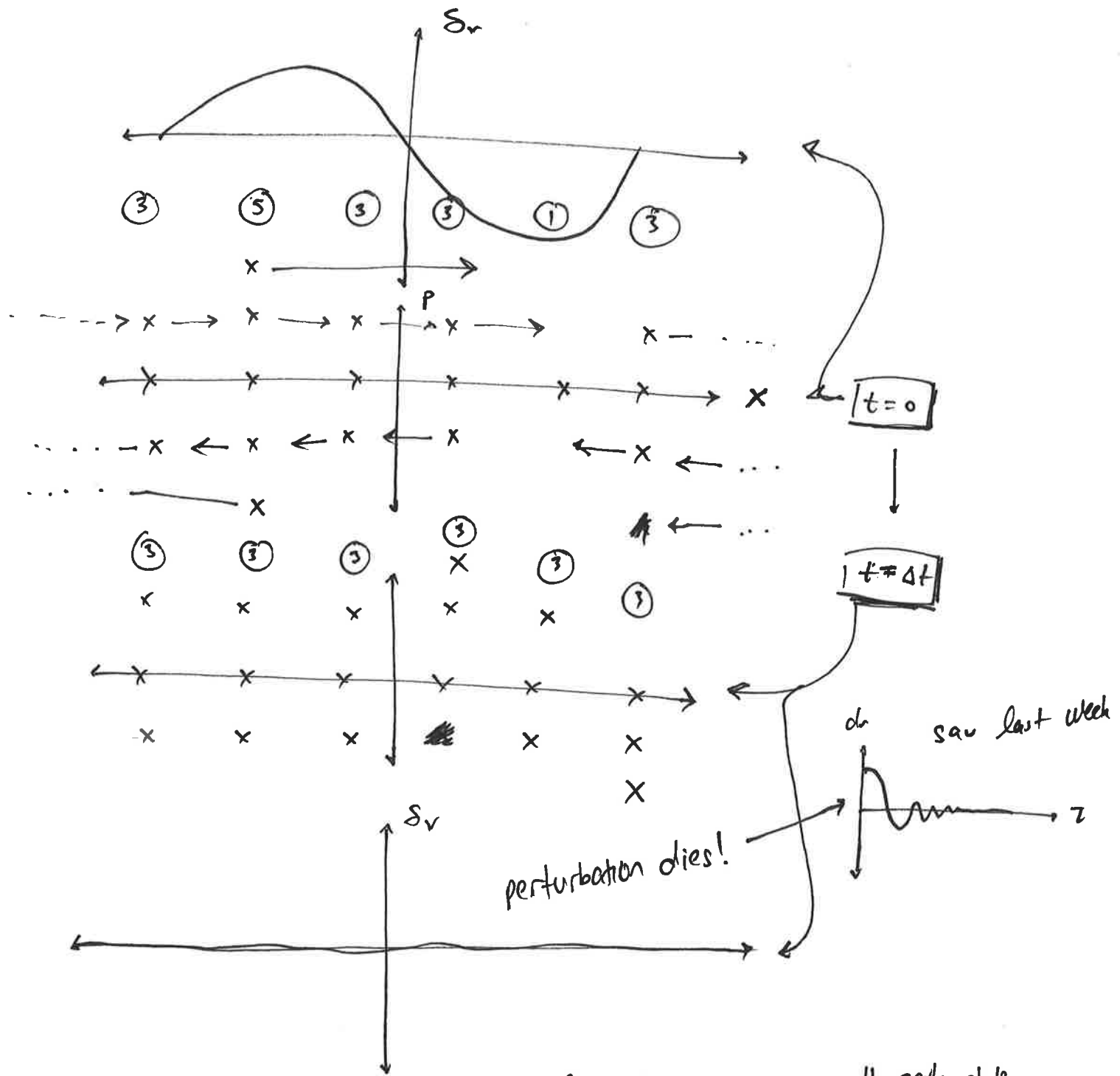
Switch from solving TCA eqn to solving full eqn once
 $T_{\text{slow}} \sim 10\% T_{\text{fast}}$

You can't get away w/ just naive TCA.

↳ show all those other terms.

In particular silk damping

To see: let's first think about v (e.s. $\sigma_T \rightarrow 0$)



This is $\lambda_{mp} \rightarrow \infty$ limit BUT

for finite λ_{mp}
 $\lambda < \lambda_{mp}$ damped.
 $k \gg 1/\lambda_{mp}$

small scale photo
 pert damped b/c
 v free streaming.